



BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals



Research Tip: Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

- It is an organised collection of information that is stored electronically so that it can be easily accessed, managed, updated & retrieved
- It is essential in the modern world because businesses, schools, hospitals, banks etc. handle huge amount of info. every day & databases makes it possible to store the info. securely.
- Banks use database to store customer details, account balances, transaction history etc.



- Hospital uses databases to store patient records, including personal information, medical history, diagnoses, test results & treatment info.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

- Data warehouse is a centralized system designed to collect, store & analyze large amount of data from diff sources
- It differs - ~~It integrates~~ It analyses data & produce insights
- Company will use it because - it brings the information together so the company can ask questions such as:
 * How much revenue did we generate each month.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

- Data lake is a centralized storage system that allows an org to store large amount of raw data.
- * Data lake does not require data to be cleaned & structured before it is stored.
- * They would prefer data lake to store very large amount of dif. types of data.
- * It keep data in its raw form for future use

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

- It is a smaller, focused collection of data designed for particular dept., business function or group of users
- * The access is given per departmental use.
- How it relate to Data warehouse.
 * Data sources → Data warehouse → Data marts → users/data tools

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

- Structured** - Data is stored organised in a fixed format, usually in Rows Columns making it easy to analyze.
- Unstructured** - Data does not have predefined structure or consistent format - making it more difficult to analyze.



- ④ ML-models, particularly deep neural networks such as convolutional neural networks (CNN) & vision transformers learn patterns from large collections of images
- ⑤ Interpretation - The system uses these learned patterns to identify what is present
- The Real world use Cases

Self driving cars - Computer vision helps detect roads, traffic signs, pedestrians, other vehicles & obstacles so the car navigates safely

medical image: Computer vision can analyze X-ray CT scans, & MRI images to help Doctors detect abnormalities such as fractures or tumors

1.3. Computer vision that a computer converts visual data into an understood information & act upon



Section C: APIs, Tools & Developer Concepts

Research Tip: Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.



6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

- is the process of organizing → representing data, its relationships & the rules that govern how data is stored & used in the system.
- Important:
 - It organizes data.
 - reduces duplicates.
 - improves data accuracy
 - makes system easier to maintain.

Models

- ① Hierarchical Data model
 - It organizes data in a tree like structure.
- ② Relational Data model
 - It organizes data into tables consisting of two rows.

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

- It is the process of examining large amount of data to discover useful patterns, relationships, trends & insights.
- It is used to understand customers - buying habits.
 - predict future trends - changes in demand or customer behavior
 - improve marketing - by identifying the right customers for specific product.
 - Detect fraud - finding unusual transactions
 - manage inventory - predict which product is likely to sell more
 - improve decision making - using patterns found from historical business data.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

- DBMS - It is a software that allows users & applications to create, store, organize, retrieve, update & manage data in a database.
- Popular DBMS:
- MySQL - an open-source relational DBMS that uses SQL
 - PostgreSQL - Excellent for complex, reliable & data intensive applications
 - Oracle - suited to large-scale enterprise → mission-critical systems
 - MongoDB - well suited to flexible, semi-structured & rapidly changing data.

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

- It is a data integration process used to collect data from different sources, clean & organize it & then store it where it can be analysed.
- ① Extract - Collecting data from one source or more
 - eg. Company extracts customer info from its CRM
 - ~~3. transform~~ sales transactions from its database → website analytics.
- ② Transform - Cleaning & converting the extracted data into a consistent useful format.
 - eg. If one system record dates as 14/09/2026 & another as 2026/09/14, the transformation process can convert both into the same format.
- ③ Load - putting the transformed data into a destination system, where it can be stored or analysed.



10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

CSV (Comma separated values) - a simple human-readable text format where data is stored in rows.

JSON (Javascript object notation) - text based, semi-structured format that stores data key-value.

Use it when sharing relatively small datasets, exporting/importing data between simple tools.

Use it when working with API responses, logs, or data whose structure varies between records.

Avro - Row oriented, compact binary format designed for big data analytics. Use when data is being streamed or exchanged between systems such as with Kafka, where records are typically processed one at a time.

Parquet - A Columnar - compressed storage format designed for big data analytics.

Comparison.

Use CSV for simplicity
JSON - for flexible/nested data
Avro - for streaming → schema managed data exchange
Parquet - for efficient large scale analytical storage

Question 9

ETL.

Extract Sources

- my SQL
- Excel
- web services
- Cloud

(2) Transform.

- * Remove duplicates
- * Handle missing / incorrect values
- * Combine data from diff sources
- * Calculate new fields.
- * Filtering out irrelevant data

(3) Load →

- * Data warehouses
- * Data lakes
- * Databases.
- * Cloud analytical platform →

ETL = helps improve data quality
clearing errors, duplicates, inconsistencies —
making analysis more reliable.

Question 13
AT ML

DL

* for recognising a cat in an image —
AI might rely on manually designed features
machine learning can learn characteristics
of cats from examples, while deep learning
can automatically learn features such
as edges, shapes, eyes & ears through
multiple neural network layers

How do Neural Network work
— Each neuron receives input, assigns different
weights to them, adds a bias & passes the
result through an activation function



Section B: Artificial Intelligence & Machine Learning

Research Tip: Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

— It is a branch of Computer Science that focuses on creating machines & computer systems that can perform tasks that normally require human intelligence.

Narrow AI (Weak AI) — designed to perform limited task e.g Siri on iPhone. Can understand spoken commands, answer questions & perform certain actions but it does not possess general human intelligence.

General AI (Strong AI) — refers to hypothetical form of AI that will have many different tools & functions like: intelligence across many different tasks & domains e.g. hypothetical form of AI that would learn math, write a novel, diagnose mechanical problems.

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

— machine learning (ML) is a branch of AI that enables computers to learn from data & improve their performance without being explicitly programmed for every task.

① **Supervised** — The model is trained using labelled data — correct answers are provided during training. — The system learns the relationship between input & output.

② **Unsupervised** — is trained using unlabelled data — no predefined correct answers. — The system finds hidden patterns, groups within the data.

③ **Reinforcement** — An agent learns by interacting with its environment. — it receives rewards for desirable actions & penalties for undesirable ones.

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

— DL is a branch of machine learning (ML) that uses artificial neural networks with many layers to learn complex patterns from large amounts of data.

Relation to AI ML DL

— The broad field of creating machines that can perform tasks requiring human intelligence.

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

— NLP is a branch of AI that enables computers to understand & interpret, process & generate human language.

— Using English



The Real world applications of NLP

- voice assistants & apps such as Siri, Google assistant & Alexa uses NLP to understand queries & respond.
- Predictive text & auto correct - messenger apps & keyboards to predict the next word.
- Email Spam filters - Email uses NLP to analyze the content of msg & identify unwanted emails, separate helpful & useful.

15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

- LLM - is a type of artificial intelligence model trained on very large amount of text to understand & generate human language.
- How a standard ML model is usually designed & trained for a specific task such as predicting house prices, detecting spam emails or classifying images.
- ① GPT (Generative pre-trained transformer) - developed by open AI - It is used for answering questions, writing & editing text, reasoning, summarization & code generation.
- ② Gemini - developed by Google - used for conversational AI & same as GPT.

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

- Generative AI - is a type of artificial intelligence that can create new content, such as text, images, music & videos based on prompts from user.
- On high level - GAI are trained on very large amounts of existing data.
- Tools: CHATGPT - Generates & understands text. It answers & summarizes info. help with coding.
- DALL-E - Generate images from written descriptions, such as creating illustrations of a robot library.
- Codex Copilot - Uses generative AI to help programmers write code, suggest code completion, explain code.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

- Training Dataset - is a collection of examples used to teach an AI model how to perform a task.
- Why it is important: - The quality of the training data strongly affects the quality of the AI model. - AI can only learn from the info it is given.
- What could go wrong if training data is biased - the model may learn and reproduce that bias. e.g. if a facial recognition system is trained mostly on images of one demographic group, it may perform better for that group & worse for others.

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

Computer vision - It is a branch of AI that enables computers to analyze, understand & interpret visual info such as images & videos.

- How do machines "see" & interpret images?
 - Image capture - on camera / sensor captures an image
 - Pixel representation - The image is converted into pixels where each pixel represented by numerical values for colour & brightness.
 - Feature extraction - Computer vision algorithms, identify patterns such as edges, shapes, textures, colours & objects.



33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

Data analyst - understand data - analyse trends, create reports
 Data Scientist - predict & explain - Build statistical/ml models
 find patterns.
 Data Engineer - Build data infrastructure - Collect, transform, store & deliver reliable data
 AI Engineer - Build AI-powered apps - integrate/train/deploy AI-models into real product.

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

- Refers to the use of technologies, tools & methods to collect, analyse & present business data
 1. Microsoft Power BI - help businesses to connect data from different sources.
 2. Tableau - Visualise & analyse data through interactive charts, graphs & dashboards.

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

→ Data modelling - in the many environment if you are equipped with the skill it will make your job easier and collect useful data & store it accordingly

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?



Section D: Data Ethics, Governance & Careers

Research Tip: Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.



I found the AI concept broad and complex but after relating it to my day to day activities it really opened my eyes.
 - Examples received from CHATGPT - made me understand how data is connected

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

- It will have major effect on my future career & everyday life
 In my career, I expect it to help me work faster.
 In the future - lot of people are going to be replaced because only few people will be required to do the job currently done by 50 people

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. CHATGPT - it is the only source I used.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.

Good luck! We believe in you.

BrightLearn Data Analytics

